

Homework B1 | ECON 0100 | Fall 2026

Due: Sunday, Sep. 20

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Demanded

Use a graph to plot this demand curve, including the quantity demanded at both 40 galleons and 50 galleons.

- a) Quantity demanded at 40 galleons: _____
- b) Quantity demanded at 50 galleons: _____
- c) How much did quantity demanded change as the price dropped from 50 to 40? _____
- d) What is the marginal benefit at a quantity of 100? _____

Q2 | Consumer Surplus

Then find and label the consumer surplus at these prices.

a) CS at 40 galleons: _____

b) CS at 50 galleons: _____

c) How much did consumer surplus change as the price dropped from 50 to 40? _____